

# 数学建模的优化方法与案例分析

几种常见连续优化方法与正则化方法简介

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中南大学数学建模竞赛培训

# 我的职业路径

- 2022.04–now Associate professor, Central South University, Changsha, China.
- 2019.10–2022.03 Research scientist, joint appointment by Humboldt University (HU) of Berlin (75%) and Weierstrass Institute for Applied Analysis and Stochastics (WIAS) in Berlin (25%), Germany.
- 2017.09–2019.09 Research assistant (Postdoc), Institute of Mathematics, HU-Berlin, Germany.
- 2012.10–2017.08 Research assistant (PhD), Computational Science Center, University of Vienna, Austria.
- 2010.09–2011.03 Exchange master student, Kuopio campus, University of Eastern Finland.
- 2007.07–2012.09 Secretary of scientific affairs, Faculty Math. & Comp. Sci., Hunan Normal University, China.
- 2003.09–2007.06 Undergraduate student, Faculty Math. & Comp. Sci., Hunan Normal University, China.

# 数学建模的一般步骤

一般数学建模的过程通常涉及以下步骤：

- 定义目标
- 查找相关文献
- 建立模型并收集数据
- 初始测试
- 验证模型

这个过程可能需要重复地执行！

# 本次课程涵盖的主要内容

涉及的内容:

- 连续优化中的一些基本方法
- 针对病态问题的正则化方法
- 变分法与偏微分方程数值方法简介
- 图像科学中的优化与正则化问题

# Outline

- 1 连续优化方法
- 2 正则化方法
- 3 变分法与偏微分方程数值方法简介
- 4 图像科学中的优化与正则化方法

# 什么是连续优化问题

连续优化问题是指决策变量所在的可行集合是连续的，如：

$$\min_{x \in X} f(x) \quad (1)$$

这里的可行集  $X$  可以是数域、抽象空间、平面、连续集合等。

当  $X$  是一些离散集合时，如整数集合、离散点集等，我们称(1)为离散优化问题。离散优化是一大类非常重要的问题，无论是在比赛中，还是在实际应用领域，均广泛存在。

本课程主要专注于连续优化问题，这与现代分析学、变分学、科学计算、大数据与机器学习等热门研究领域息息相关。

# 如何理解连续优化问题的最优解

继续考虑优化模型

$$\min_{x \in X} f(x)$$

一个重要而基本的问题

- 该问题的最优解应该如何刻画?
- 或者说, 该如何求解该连续优化问题?

这取决于可行集  $X$  和函数  $f$  的性质 (光滑性、凸性等)。  
对于全空间上的可微函数, 我们可考虑费马极值原理

$$x^* \text{ minimizes } f \quad \Rightarrow \quad \nabla f(x^*) = 0.$$

# 如何理解连续优化问题的最优解

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对于  $X$  是子集的情形, 即带约束优化问题, 我们可考虑如下不等式条件

$$x^* \text{ minimizes } f \quad \Rightarrow \quad \langle \nabla f(x^*), x - x^* \rangle \geq 0 \quad \text{任意的 } x \in X.$$

上述不等式被称为变分不等式。



# 最小二乘问题

最小二乘是数学建模过程中极其常用的一个优化模型。

给定一组数据  $(x_i, y_i)_{i=1}^n$ ，对于每个数据对  $(x_i, y_i)$ ，我们假定有一个参数模型  $r_\theta(x_i) \sim y_i$ ，这里  $\theta \in \mathbb{R}^m$  是一些未知参数。通过确定这些未知参数，我们可得到该参数模型  $r_\theta$  的具体形式，从而对新的取样  $x$  进行预测。这也是机器学习  
中所谓监督学习的思想。

如何确定这些未知参数  $\theta$  是我们希望解决的问题。

借助最优化的思想，我们可考虑求解如下问题：

$$\min_{\theta \in \mathbb{R}^m} \frac{1}{n} \sum_{i=1}^n |r_\theta(x_i) - y_i|^2,$$

此即最小二乘问题（分为线性与非线性情形）。

最小二乘问题可以解决数据不足或数据过载等情形（分别对应于线性方程组欠定或超定情形），但无法克服病态性问题。

# 梯度下降法和梯度流方程

- Gradient flow ([Showalter's method in Inverse Problems](#)):

$$\dot{u}(t) = -\nabla f(u(t)) \quad \text{with } u(0) = u_0 \in U$$

- Gradient descent ([Landweber iteration in Inverse Problems](#)):

$$u_k = u_{k-1} - \alpha \nabla f(u_{k-1}) \quad \text{giving some } u_0 \in U$$

# 加速梯度法与二阶动力学

## Discrete setting

- Momentum method <sup>1</sup>:

$$\begin{aligned}u_k &= y_{k-1} - \alpha \nabla f(y_{k-1}) \\ y_k &= u_k + \beta(u_k - u_{k-1}) \text{ given some } u_0, u_1 \in U\end{aligned}\tag{2}$$

- Nesterov's method <sup>2</sup>:

$$\begin{aligned}u_k &= y_{k-1} - \alpha \nabla f(y_{k-1}) \\ y_k &= u_k + \frac{k-1}{k+2}(u_k - u_{k-1}) \text{ given some } u_0, u_1 \in U\end{aligned}\tag{3}$$

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<sup>1</sup>Polyak. Some methods of speeding up the convergence of iteration methods. USSR Computational Mathematics and Mathematical Physics, 1964.

<sup>2</sup>Nesterov. A method for unconstrained convex minimization problem with the rate of convergence  $o(1/k^2)$ . Doklady ANSSSR (translated as Soviet.Math.Docl.), 1983.

# 加速梯度法与二阶动力学

## Continuous setting

- Heavy ball dynamics <sup>1</sup>:

$$\ddot{u}(t) + \eta \dot{u}(t) = -\nabla f(u(t)) \quad \text{and} \quad \dot{u}(0) = v_0, \quad u(0) = u_0$$

- Viscosity vanishing flow <sup>2</sup>:

$$\ddot{u}(t) + \frac{\eta}{t} \dot{u}(t) = -\nabla f(u(t)) \quad \text{and} \quad \dot{u}(0) = v_0, \quad u(0) = u_0$$

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<sup>1</sup>Attouch, Goudou and Redont. The heavy ball with friction method. I. The continuous dynamical system, Communications in Contemporary Mathematics, 2000

<sup>2</sup>Su, Boyd, and Candes. A differential equation for modeling Nesterov's accelerated gradient method: theory and insights. The Journal of Machine Learning Research, 2016.

# 牛顿法与拟牛顿法

- Newton's method:

$$u_k = u_{k-1} - (\nabla^2 f(u_{k-1}))^{-1} \nabla f(u_{k-1}) \text{ giving some } u_0 \in U$$

$(\nabla^2 f(u_{k-1}))^{-1}$  is called Hessian, which needs the second-order derivatives of  $f$ .

- Many methods are invented to replace the Hessian by approximation. These type of methods are usually called quasi-Newton methods.

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- ① 连续优化方法
- ② 正则化方法
- ③ 变分法与偏微分方程数值方法简介
- ④ 图像科学中的优化与正则化方法

# 什么是正则化方法

考虑一病态矩阵方程

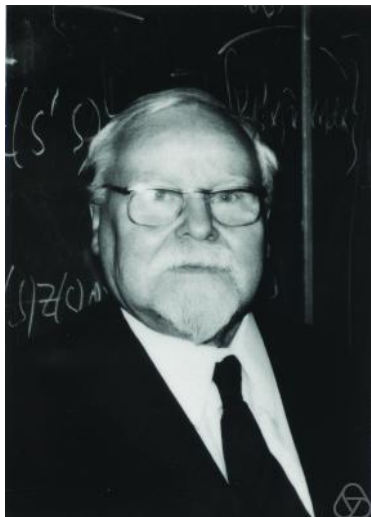
$$Ax = y$$

where

$$A = \begin{bmatrix} 2^{-1} & 0 & 0 & \cdots & 0 \\ 0 & 2^{-2} & 0 & \cdots & 0 \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & 0 & 0 & \ddots & 0 \\ 0 & 0 & 0 & \cdots & 2^{-n} \end{bmatrix}$$

What gonna happen if there is some noise in the data  $y^\delta = y + \delta$ ?

## Tikhonov 正则化方法<sup>3</sup>



Andrey Nikolayevich Tikhonov proposed the following method:

$$u(\alpha, y^\delta) := \arg \min_{u \in U} \|Au - y^\delta\|_Y^2 + \alpha \|u\|_U^2$$

to approximate the solution of the linear equation.  $\alpha > 0$  is called a regularization parameter.

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<sup>3</sup>Tikhonov, A. N. (Translated) "Solution of incorrectly formulated problems and the regularization method". Soviet Mathematics. 4: 1035–1038, 1963.



# Landweber 正则化方法

# 动力学正则化方法

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# Dirichlet 能量极小化与 Laplace 方程

The following functional is called Dirichlet energy

$$\min_{u \in H^1(\Omega)} \frac{1}{2} \|\nabla u\|_{L^2(\Omega)}^2$$

It turns out that the minimizer will satisfy the following partial differential equation

$$-\Delta u = 0 \text{ in } \Omega \quad \text{with} \quad \partial_\nu u = 0 \text{ on } \Omega.$$

# 热方程与梯度下降法

$$\min_{u \in H^1(\Omega)} \frac{1}{2} \|\nabla u\|_{L^2(\Omega)}^2$$

Derive the gradient flow of Dirichlet energy, that becomes

$$\partial_t u = \Delta u \text{ in } \Omega \times (0, \infty).$$

# 偏微分方程数值方法

Some of the current methods

- Finite difference methods (based on the strong form of the PDEs)
- Galerkin methods (based on the weak form of the PDEs), including finite element methods, spectral methods, among the others
- Machine learning methods

# 动态微分方程数值方法

Let us consider the following abstract equation:

$$\partial_t u = F(u).$$

Three ways to deal with the temporal discretization

- Taylor expansion
- Finite difference formulas
- Integration formulas

Spatial discretization has been mentioned earlier.

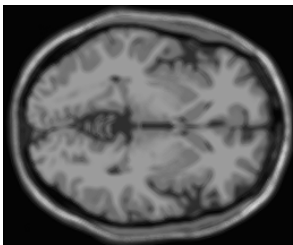
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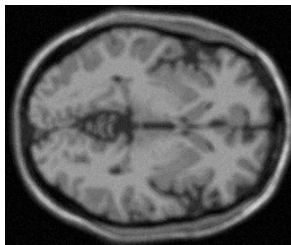


## 例题解析

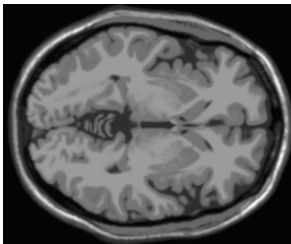
Recover image from blurred noisy data:  $y^\delta = k * u + \delta$



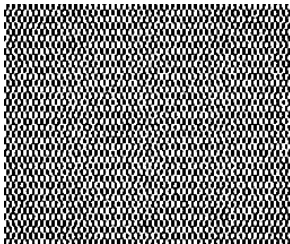
Blur without noise



Blur with noise



Reconstruction



Reconstruction (fails)

# Questions?

Thank you for your attention!

Any questions?

Contact: [guozhi.dong@csu.edu.cn](mailto:guozhi.dong@csu.edu.cn)